Digital Trace Data

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Digital Trace Data

Every time we interact with the internet we leave behind fingerprints

These data are what we call 'digital traces.'

Some of these data are more obvious to us: If we post on TikTok, reddit, X, Bluesky, or elsewhere, we should anticipate that post is now public data. Some we might perceive to be private: (private) Instagram posts, WhatsApp or Signal messages, etc. Yet others we aren't even aware are being collected.

A rough taxonomy:

- **Social media activity** (e.g., Twitter/X, Reddit, TikTok, Bluesky)
- **Mobility traces** (e.g., smartphone GPS; telecom mobility indices / aggregates)
- **Information seeking** (e.g., Google Trends; Wikipedia page views)
- **Information consumption** (e.g., Comscore or YouGov panels; browsing or app usage logs)

Overview

1. Cellular tracking and mobility data

2. Internet search data

3. Wikipedia

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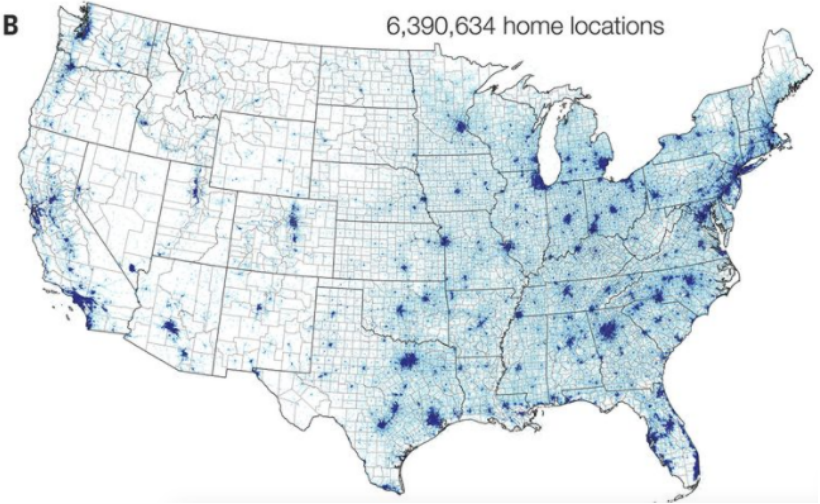
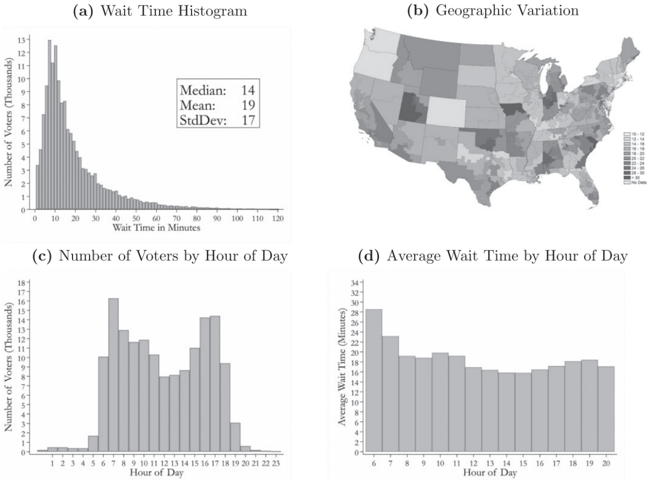
Chen & Rohla 2018



Chen, M. K., & Rohla, R. (2018). The effect of partisanship and political advertising on close family ties. *Science*, 360(6392), 1020-1024.

Chen et al. 2022

FIGURE 2.—OVERALL WAIT TIMES

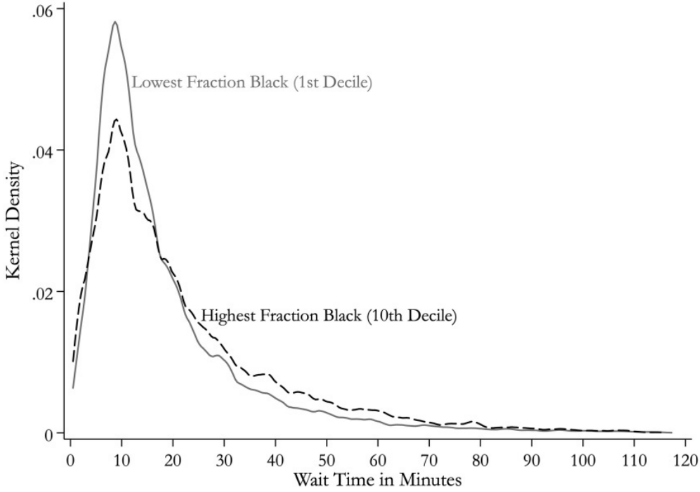


Panel A plots a histogram corresponding to the 154,495 cell phones that pass the filters used to identify likely voters (using 1.5 minute bins). Panel B shows variation in (empirical-Bayes-adjusted) average wait times by congressional district (115th Congress). Panel C plots the total number of voters (volume) by hour of arrival. Panel D plots the average wait time for each hour of arrival.

Chen, M. K., Haggag, K., Pope, D. G., & Rohla, R. (2022). Racial disparities in voting wait times: Evidence from smartphone data. *Review of Economics and Statistics*, 104(6), 1341-1350.

Chen et al. 2022

FIGURE 3.—WAIT TIME: FRACTION BLACK FIRST VERSUS TENTH DECILE



Kernel densities are estimated using 1 minute half-widths. The first decile corresponds to the 34,420 voters across 10,319 polling places with the lowest percent of black residents (mean = 0%). The tenth decile corresponds to the 15,439 voters across the 5,262 polling places with the highest percent of black residents (mean = 58%).

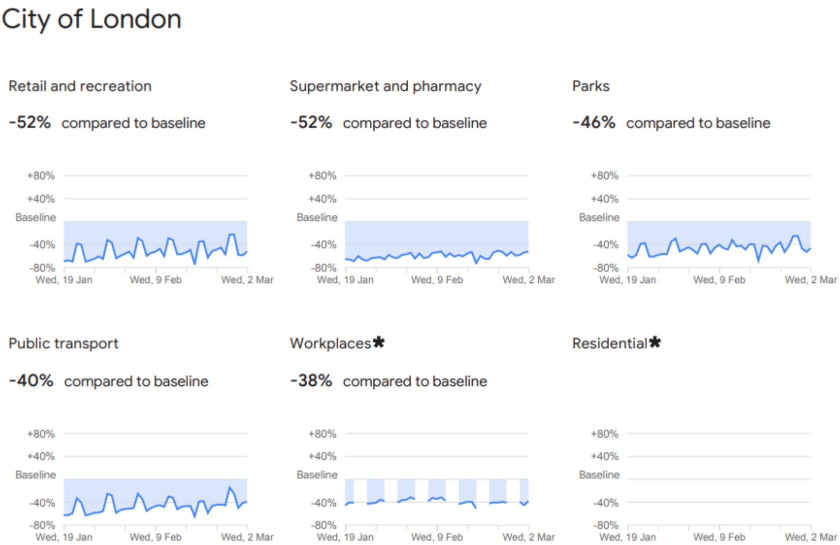
Google Mobility

See how your community is moving around differently due to COVID-19

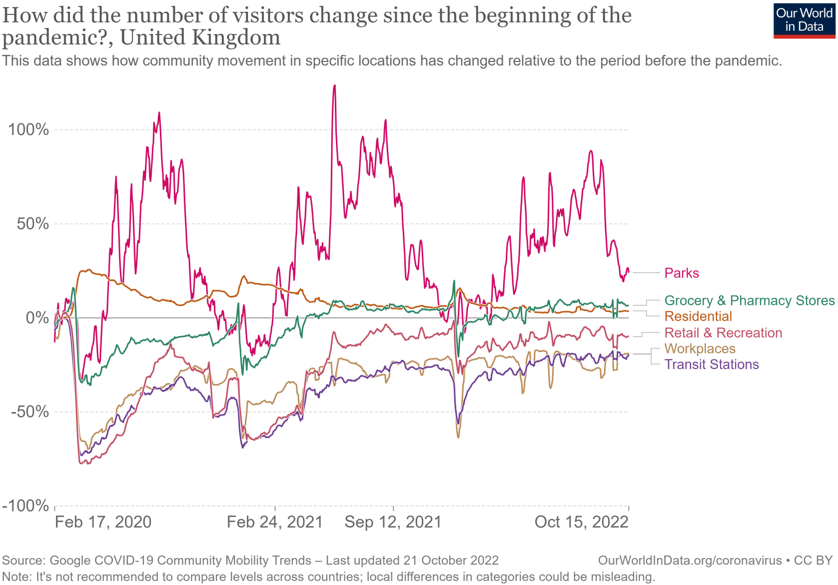
As global communities respond to COVID-19, we've heard from public health officials that the same type of aggregated, anonymized insights we use in products such as Google Maps could be helpful as they make critical decisions to combat COVID-19.

These Community Mobility Reports aim to provide insights into what has changed in response to policies aimed at combating COVID-19. The reports chart movement trends over time by geography, across different categories of places such as retail and recreation, groceries and pharmacies, parks, transit stations, workplaces, and residential.

Google Mobility - City of London



Google Mobility - UK



Ansell, Cansunar, & Elkjaer 2021

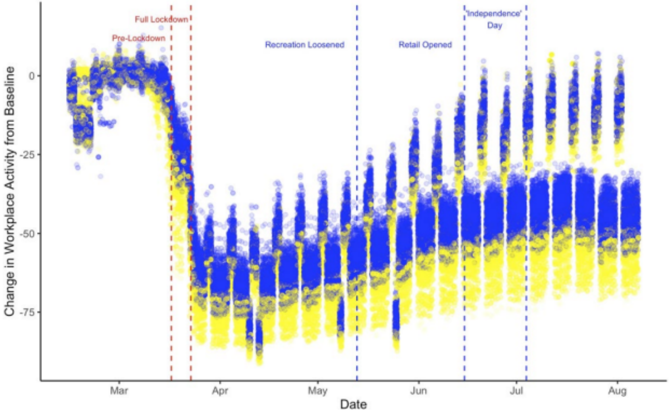


Figure 1. UK workplace activity, government announcements, and the Brexit vote. Note: Blue: Voted 'Leave'; Yellow: Voted Remain.

Ben Ansell, Asli Cansunar & Mads Andreas Elkjaer (2021) Social distancing, politics and wealth, *West European Politics*, 44:5-6,

Mobility / location data

Ethical and privacy considerations

- Legal limitations on access to precise location data
- Difficulty of fully anonymizing precise location data
- Difficulty of protecting “group privacy” (e.g., military personnel)
- Representativeness (e.g., opt-in apps tend to be biased towards the more affluent)
- Purpose limitation (e.g., data collected in a crisis could be repurposed by governments for other uses)

Overview

Internet search data

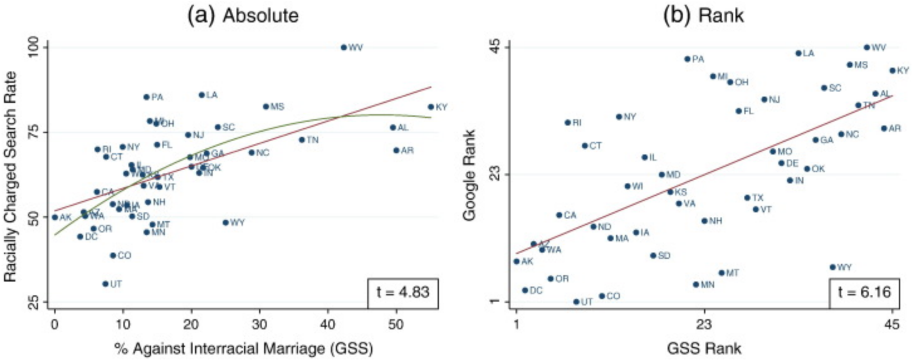
People's "true" opinions and beliefs are difficult to measure $\longrightarrow$ social desirability bias

Methods for studying sensitive topics: IAT, list experiments

More recently, what people search for as an (anonymized) proxy for private or taboo beliefs and behaviours

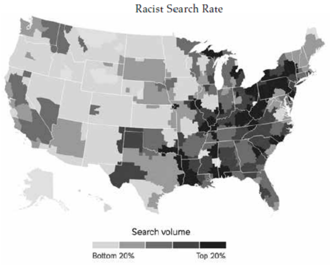
Be aware of biases and conceptual issues:

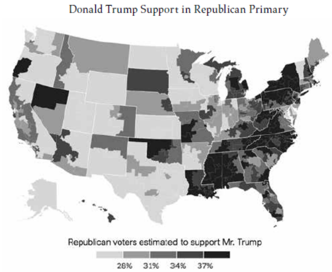
- Who are searches representative of?
- Is searching the same as believing something?
- Is searching the same as actually doing something?



Stephens-Davidowitz, S. (2014). The cost of racial animus on a black candidate: Evidence using Google search data. *Journal of Public Economics*, 118, 26-40.

Stephens-Davidowitz 2017





Krupenkin et al 2024

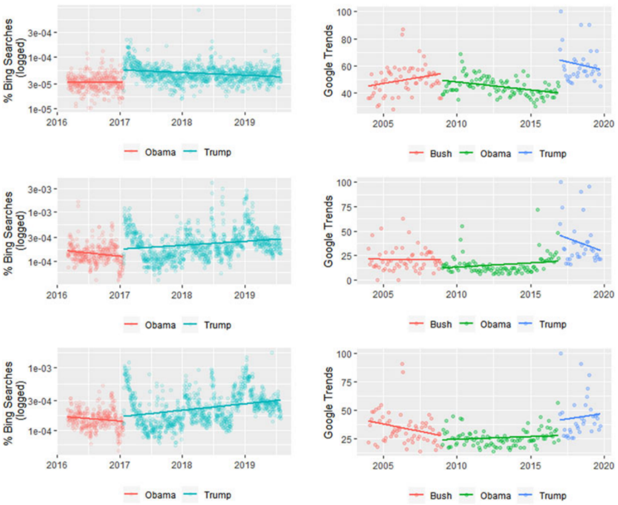
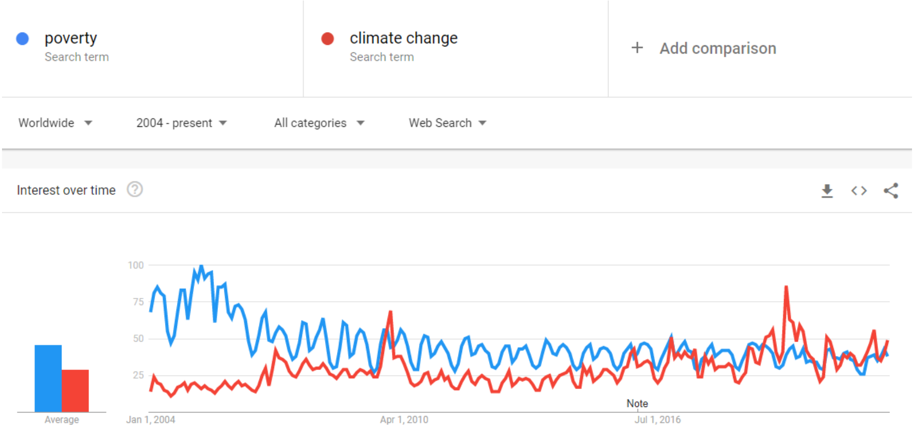


Figure 4. Immigration searches by administration.
Notes: For all three sets of anti-immigrant terms, searches increased on Bing and Google after Trump's inauguration speech. There was also a spike in all three sets of terms in the months immediately after the inauguration. For the Bing data, the *Y*-axis is daily searches for the immigration terms as a percentage of daily Bing searches on a log10 scale, and each data point represents one day. For the Google data, the *Y*-axis is Google Trends' 'Interest over Time' measure, and each data point represents one month.

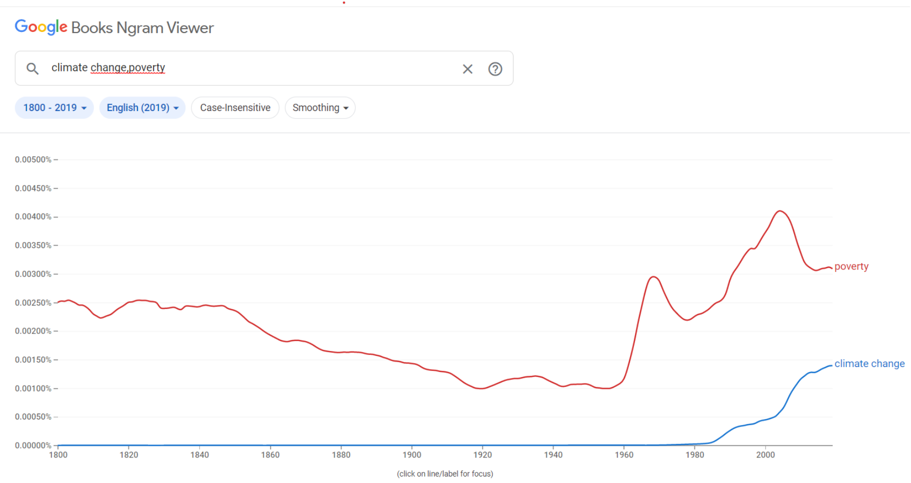
Google trends example



See `https://trends.google.com/trends/?geo=GB`

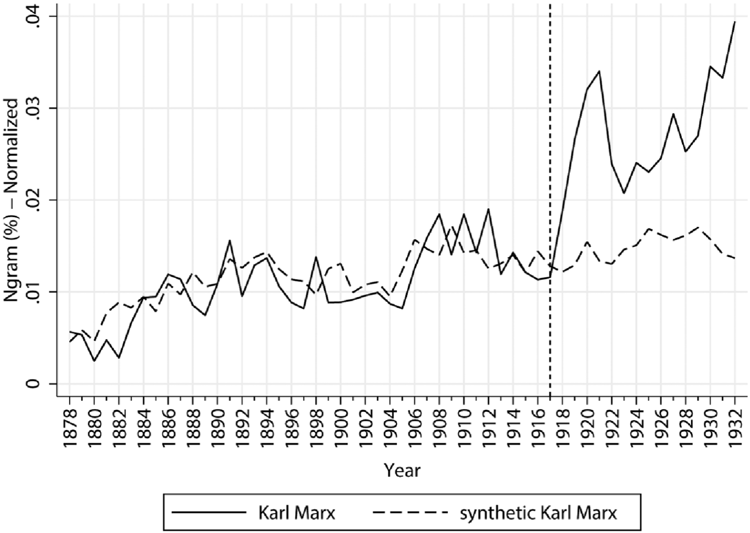
For lessons: `https://newsinitiative.withgoogle.com/resources/products/google-trends/`

Google Ngrams example



See `https://books.google.com/ngrams/info`

Magness & Makovi 2022



Magness, P. W., & Makovi, M. (2023). The Mainstreaming of Marx: Measuring the effect of the Russian Revolution on Karl Marx's influence. *Journal of Political Economy*, 131(6), 1507-1545.

Overview

Wikipedia

Wikipedia data

- "Online encyclopedia written and maintained by a community of volunteers through a model of open collaboration, using a wiki-based editing system"
- Raw and open data, i.e., not controlled and pre-processed like Google search data
- Accessible via API. See R package WikipediR
- Limitation: can't differentiate by country
- Be aware of biases, e.g., Wikipedia editors are not representative of the population

Wikipedia data: Applications to politics research

- "Edit wars" and corrections (e.g., Gobel & Munzert 2018, Kalla & Aranow 2015)
- Bias in Wikipedia articles (e.g., Greenstein & Zhu 2012, Pradel 2021)
- Page views as an indicator of interest (e.g., Atkinson & DeWitt 2019)
- Page views as an indicator of support (e.g., Yasseri & Bright 2016, Salem & Stephany 2021)
- Censorship (Pan & Roberts 2020, Hobbs & Roberts 2018)
- Clickstream data (e.g., Gessler *unpublished*)

Example: Pradel 2021

- Do search engine suggestions and Wikipedia entries for politicians differ based on the gender or party of the politician?
  - Personal (e.g., family, friends, leisure, body, sexuality) versus role oriented (e.g., work, political parties and organizations, political events) information

Franziska Pradel (2021) Biased Representation of Politicians in Google and Wikipedia Search? The Joint Effect of Party Identity, Gender Identity and Elections, *Political Communication*, 38:4, 447-478

Pradel 2018

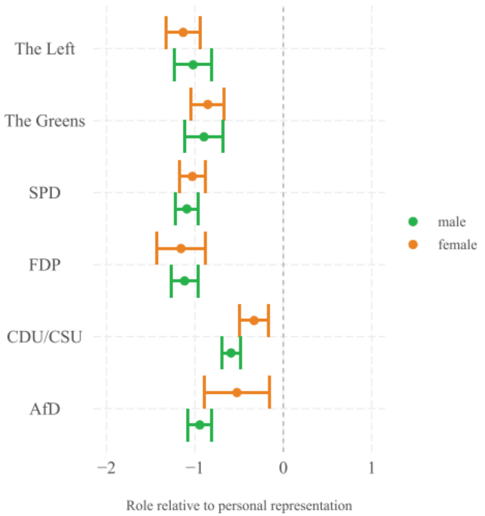


Figure 3. Predicted effects for personal-role information by gender and party identity (ordered from left-wing to right-wing parties).